

AlgoChowk — AI Usage Note

Tools used

ChatGPT was used as an AI coding and documentation assistant while preparing the research notebook and submission documents.

How AI was used

AI was used to structure the Python notebook, draft data-validation logic, implement event detection and forward-return calculations, create configurable robustness checks, add an out-of-sample split, and implement a simple event-driven backtest. AI was also used to draft the README and this research note.

My own research decisions

The research design uses a -2% one-day fall as the default event threshold, next-day Open as the entry price, 3/5/10 trading-day holding periods, and a simple 5-day event-driven backtest. The notebook keeps these assumptions configurable rather than hard-coding a single optimised result.

Suggestions changed or treated cautiously

The implementation was structured around the assignment's emphasis on avoiding look-ahead bias, comparing against a baseline, testing robustness and validating out of sample. Historical performance is not treated as sufficient evidence of a trading edge.

AI limitations / checks

AI-generated code can contain implementation errors and can encourage overfitting if parameters are selected after looking at results. The notebook therefore exposes assumptions explicitly and separates development data from the final out-of-sample period. Data-source and numerical outputs should be checked by the submitter before final submission.

What I learned

The main lesson is that an event-study result needs more than an average return. Baseline comparison, uncertainty, robustness, out-of-sample validation, transaction costs and possible biases determine whether an observed relationship deserves further investigation.